

In the middle of the bridge — Solution

W24 (2024) — English translation

A1

1.50 pts

Since without a section of the circuit between nodes A and B the electrical circuit consists entirely of constant voltage sources and resistors, the dependence of the current I_{AB} flowing from node A to node B on the voltage $U_{AB} = \varphi_A - \varphi_B$ in this section of the circuit can be represented in a form that does not depend on the element connected between nodes A and B :

$$I_{AB} = \frac{\mathcal{E}_0 - U_{AB}}{r_0}.$$

Determine $\mathcal{E}_0$ and r_0 . Answers выtimesandme via/through $\mathcal{E}$ and R .

If an ideal voltmeter is connected between nodes A and B , the voltage across it is

$$U_V = \mathcal{E} \left(\frac{6R}{6R + 4R} - \frac{2R}{2R + 8R} \right) = \frac{2}{5} \mathcal{E} = \mathcal{E}_0.$$

If voltage на sourcee equals нулю, его можно замestring/thread на провод с нулевым сопротивлением. Then схема между выwaterми A and B будет представлять собой две vaporopараллельно соединенных resistorов, $(2R, 8R)$ and $(6R, 4R)$, соединенных посонsequently, and therefore resistance

$$r_0 = R_{AB} = \frac{2R \cdot 8R}{2R + 8R} + \frac{6R \cdot 4R}{6R + 4R} = 4R.$$

Then зависимость напряжения от силы currenta будет иметь вид

$$U_{AB} = \mathcal{E}_0 - I_{AB} r_0,$$

hence

$$I_{AB} = \frac{\mathcal{E}_0 - U_{AB}}{r_0}.$$

Альтернативно можно вычислить current короткого замыкания $I_c = \mathcal{E}_0/r_0$, который будет течь via/through идеальный амперметр, подключенный между выwaterми A and B . For/At таком подключении общее resistance цепи, к которой подключен source $\mathcal{E}$, equals

$$r_1 = \frac{2R \cdot 6R}{2R + 6R} + \frac{8R \cdot 4R}{8R + 4R} = \frac{25}{6} R.$$

Then полный current via/through source $\mathcal{E}$ $I_{\mathcal{E}} = 6\mathcal{E}/25R$, a currentand via/through resistorы $2R$ and $8R$ будут equal соответственно:

$$I_1 = \frac{3}{4} I_{\mathcal{E}} = \frac{9\mathcal{E}}{50R}, \quad I_2 = \frac{1}{3} I_{\mathcal{E}} = \frac{2\mathcal{E}}{25R}.$$

Then current короткого замыкания

$$I_c = I_{AB} = I_1 - I_2 = \frac{\mathcal{E}}{10R}.$$

Hence we get прежнее value сопротивления

$$r_0 = \mathcal{E}_0/I_c = 4R.$$

Answer: \textbf{Answer:}

$$r_0 = 4R, \quad \mathcal{E}_0 = \frac{2}{5} \mathcal{E}$$

A2**0.50 pts**

Obtain the dependence of the current strength I_{AB} on the value $\mathcal{E}$. Express your answer using $\mathcal{E}$, R , U_0 and I_0 .

Suppose that the voltage on the nonlinear element is less than U_0 , then this voltage can be found from the equation

$$(r_0 + R)I_{AB} = \mathcal{E}_0 - U_{AB}, \quad (r_0 + R)I_0 \frac{U_{AB}}{U_0} = \mathcal{E}_0 - U_{AB},$$

hence

$$U_{AB} = \frac{\mathcal{E}_0 U_0}{U_0 + (r_0 + R)I_0}, \quad \mathcal{E}_0 \leq U_0 + (r_0 + R)I_0.$$

Соответствующее value currenta

$$I_{AB} = \frac{\mathcal{E}_0 I_0}{U_0 + 5RI_0}, \quad \mathcal{E}_0 \leq U_0 + 5RI_0.$$

For/At дальнейшем увеличении напряжения sourcea voltage ннотлинейном элементе станет больше U_0 , a current via/through него будет equals I_0 . Voltage ннотлинейном элементе $U_{AB} = \mathcal{E}_0 - I_0 r_0$. Substituting выражения for $\mathcal{E}_0$ and r_0 , we get the answer.

Answer: \textbf{Answer:}

$$I_{AB} = \begin{cases} \frac{2\mathcal{E}I_0}{5(U_0 + 5RI_0)}, & \mathcal{E} \leq \frac{5}{2}(U_0 + 5RI_0); \\ I_0, & \mathcal{E} > \frac{5}{2}(U_0 + 5RI_0). \end{cases}$$

A3**0.50 pts**

Obtain the dependence of the power $P_{\text{н.э}}$ released on the nonlinear element on the value $\mathcal{E}$. Express your answer using $\mathcal{E}$, R , U_0 and I_0 .

Power on the nonlinear element $P_{\text{н.э.}} = U_{AB} I_{AB}$. Using the results of the previous paragraph, we get the answer.

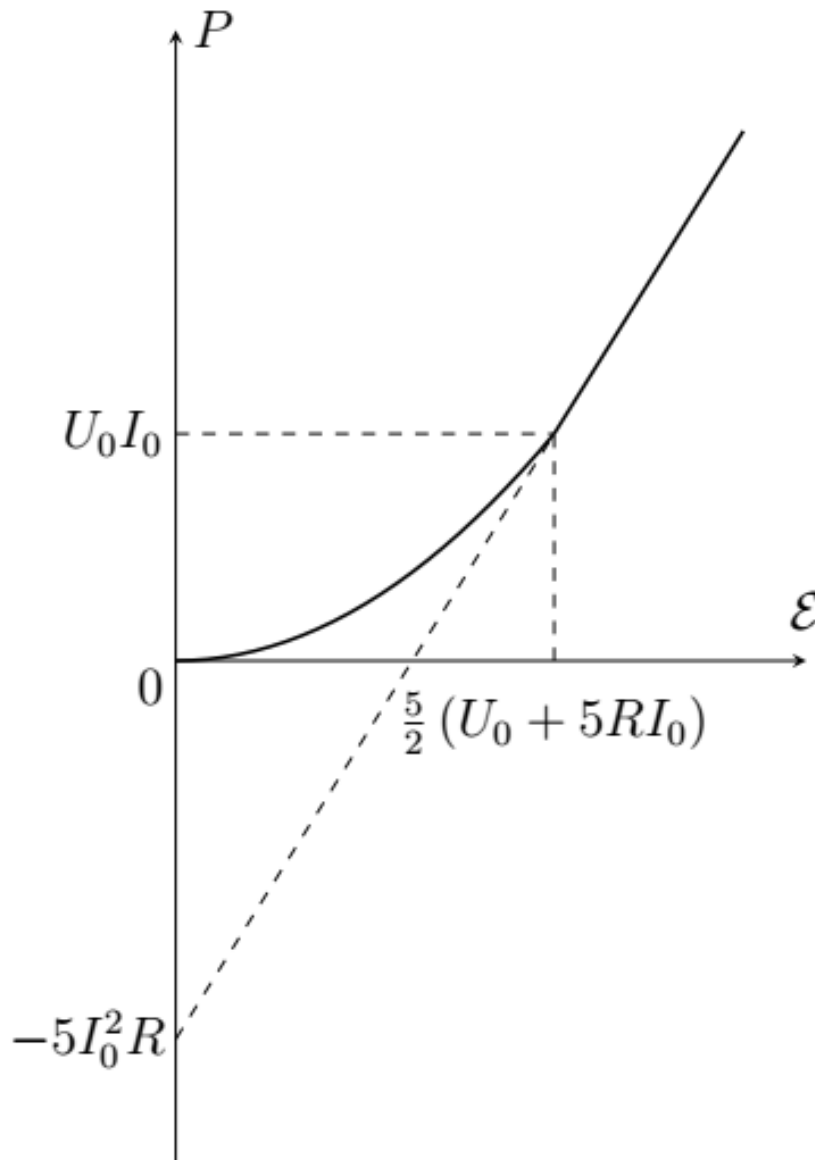
Answer: \textbf{Answer:}

$$P_{\text{н.э.}} = \begin{cases} U_0 I_0 \frac{4\mathcal{E}^2}{25(U_0 + 5RI_0)^2}, & \mathcal{E} \leq \frac{5}{2}(U_0 + 5RI_0); \\ I_0 \left(\frac{2}{5}\mathcal{E} - 5I_0 R \right), & \mathcal{E} > \frac{5}{2}(U_0 + 5RI_0). \end{cases}$$

A4**0.50 pts**

Construct a high-quality graph of the dependence $P_{\text{н.э}}(\mathcal{E})$. Indicate all the characteristic points on the graph and indicate their corresponding values.

At low voltages the graph is a parabola with the vertex at the origin, at $\mathcal{E} \leq \frac{5}{2}(U_0 + 4RI_0)$ it is a straight line. At the point at which the transition from quadratic to linear dependence occurs $\mathcal{E} = U_0 + 4RI_0$, $P_{\text{н.э.}} = I_0 U_0$.



B1

0.80 pts

Obtain the dependence of the current $I_C(t)$ through the capacitor on time t . Express your answer in terms of $\mathcal{E}$, R , U_0 , I_0 , C and t .

If the voltage across a nonlinear element is less than U_0 , it behaves like a resistor with a resistance of $r_0 = U_0/I_0 = 500 \, \Omega = R$. Therefore, in this case, the voltage across the nonlinear element must be equal to the voltage across the resistor. Then the initial voltage on the nonlinear element is equal to $\mathcal{E}/2 = 3 \, \text{V} < U_0$, so we can use the linear dependence of current on voltage everywhere. Then the charge of the capacitor and the current through it are related by the relation (which is obtained from the Kirchhoff equation)

$$I(R + r_0) + qC = \mathcal{E}.$$

Продифференцировав его по времени and taking into account that $\dot{q} = I$, we get

$$\dot{I} = -\frac{I}{C(R + r_0)}, \quad I(t) = I(0)e^{-\frac{t}{C(R+r_0)}}.$$

In initial moment времени capacitor не заряжен, therefore $I(0) = \mathcal{E}/(R + r_0)$. Then the current through the capacitor

Answer: \textbf{Answer:}

$$I_C = \frac{\mathcal{E}}{R + U_0/I_0} e^{-\frac{t}{C(R+U_0/I_0)}}$$

B2

0.60 pts

What amount of heat $Q_{\text{н.э.}}$ will be released on the nonlinear element during the establishment of constant current? Express your answer in terms of $\mathcal{E}$, R , U_0 , I_0 and C . Find the numerical value.

When charging a capacitor, a charge $q = C\mathcal{E}$ passes through the source, and it does work $A = q\mathcal{E} = C\mathcal{E}^2$. This work goes to increase the energy of the capacitor and the heat released:

$$A = \frac{C\mathcal{E}^2}{2} + Q,$$

therefore в цепи выделяется тепло $Q = C\mathcal{E}^2/2$. It is divided between the resistor and the nonlinear element in proportion to their resistances (since at the voltages in question the nonlinear element behaves like a resistor)

$$Q_{\text{н.э.}} = \frac{r_0}{r_0 + R} \frac{C\mathcal{E}^2}{2} = \frac{U_0}{U_0 + RI_0} \frac{C\mathcal{E}^2}{2} = 0.9 \text{ mJ}$$

B3

0.60 pts

After what time τ the amount of heat released on the nonlinear element will reach 80% from the value $Q_{\text{н.э.}}$? Express your answer in terms of R , U_0 , I_0 , and C . Find the numerical value.

The power on the nonlinear element depending on time has the form

$$P = I_C^2 U_0 / I_0 = A e^{-2t/t_0}, \quad A = \frac{\mathcal{E}^2 U_0 I_0}{(RI_0 + U_0)^2}, \quad t_0 = C(R + U_0/I_0).$$

Then over time t нонлинейном элементе выделяется тепло

$$Q(t) = \int_0^t P(t) dt = A \frac{t_0}{2} (1 - e^{-2t/t_0}) = Q_{\text{н.э.}} (1 - e^{-2t/t_0}).$$

At the moment времени, когда $Q = 0.8Q_{\text{AD}}$, $e^{-2t/t_0} = 0.2$,

Answer: \textbf{Answer:}

$$t = \frac{\ln 5}{2} t_0 = \frac{\ln 5}{2} C \left(R + \frac{U_0}{I_0} \right) = 0.0804 \text{ s}$$

B4

1.50 pts

Obtain the dependence of the current $I_C(t)$ through the capacitor on time t . Express your answer in terms of $\mathcal{E}$, R , U_0 , I_0 , C and t .

Now the initial voltage on the nonlinear element is greater than U_0 , which means the current through the nonlinear element (equal to the current through the capacitor) is equal to $I_C = I_0$. This current will remain constant until the voltage across the capacitor drops to U_0 . Up to this point, the charge of the capacitor depends on time as $q = I_0 t$, which means the voltage on the nonlinear element is

$$U_{\text{н.э.}} = \mathcal{E} - RI_0 - I_0 t / C.$$

Эта зависимость справедлива, пока $U_{\text{AD}} \geq U_0$, that is

$$t \leq t_1 = \frac{C}{I_0} (\mathcal{E} - U_0 - RI_0).$$

Next current будет экспоненциально затухать, как for/at обычной chargeке capacitora:

$$I = I_0 \exp \left(-\frac{t - t_1}{C(R + U_0/I_0)} \right).$$

Answer: \textbf{Answer:}

$$I_C = \begin{cases} I_0, & t \leq \frac{C}{I_0}(\mathcal{E} - U_0 - RI_0); \\ I_0 \exp \left(-\frac{t - C(\mathcal{E} - U_0 - RI_0)/I_0}{C(R + U_0/I_0)} \right), & t > \frac{C}{I_0}(\mathcal{E} - U_0 - RI_0). \end{cases}$$

B5

1.50 pts

What amount of heat $Q_{\text{н.э}}$ will be released on the nonlinear element during the establishment of constant current? Find the numerical value with an accuracy of at least 1%.

While the current remains constant, the power on the nonlinear element

$$P = I_0 U_{\text{н.э.}} = I_0(\mathcal{E} - RI_0 - I_0 t/C).$$

Then over time t_1 выделится тепло

$$Q_1 = \int_0^{t_1} dt I_0(\mathcal{E} - RI_0 - I_0 t/C) = I_0(\mathcal{E} - RI_0)t_1 - I_0^2 t_1^2/2C,$$

$$Q_1 = C(\mathcal{E} - RI_0)(\mathcal{E} - U_0 - RI_0) - C(\mathcal{E} - U_0 - RI_0)^2/2 = \frac{C}{2} ((\mathcal{E} - I_0 R)^2 - U_0^2) = 1.2 \text{ mJ}$$

Next зависимость мощности от времени имеет вид ($t' = t - t_1$)

$$P = \frac{U_0}{I_0} I_0^2 e^{-2t'/t_0}, \quad t_0 = C(R + U_0/I_0)$$

and therefore полное выделившееся тепло

$$Q_2 = \int_0^\infty dt' U_0 I_0 e^{-2t'/t_0} = \frac{1}{2} U_0 I_0 t_0 = \frac{1}{2} U_0 I_0 C(R + U_0/I_0) = 2.5 \text{ mJ}$$

Answer: \textbf{Answer:}

$$Q = Q_1 + Q_2 = \frac{C}{2} ((\mathcal{E} - I_0 R)^2 - U_0^2) + \frac{C}{2} U_0(U_0 + I_0 R) = \frac{C}{2} ((\mathcal{E} - I_0 R)^2 + U_0 I_0 R) = 3.7 \text{ mJ}$$